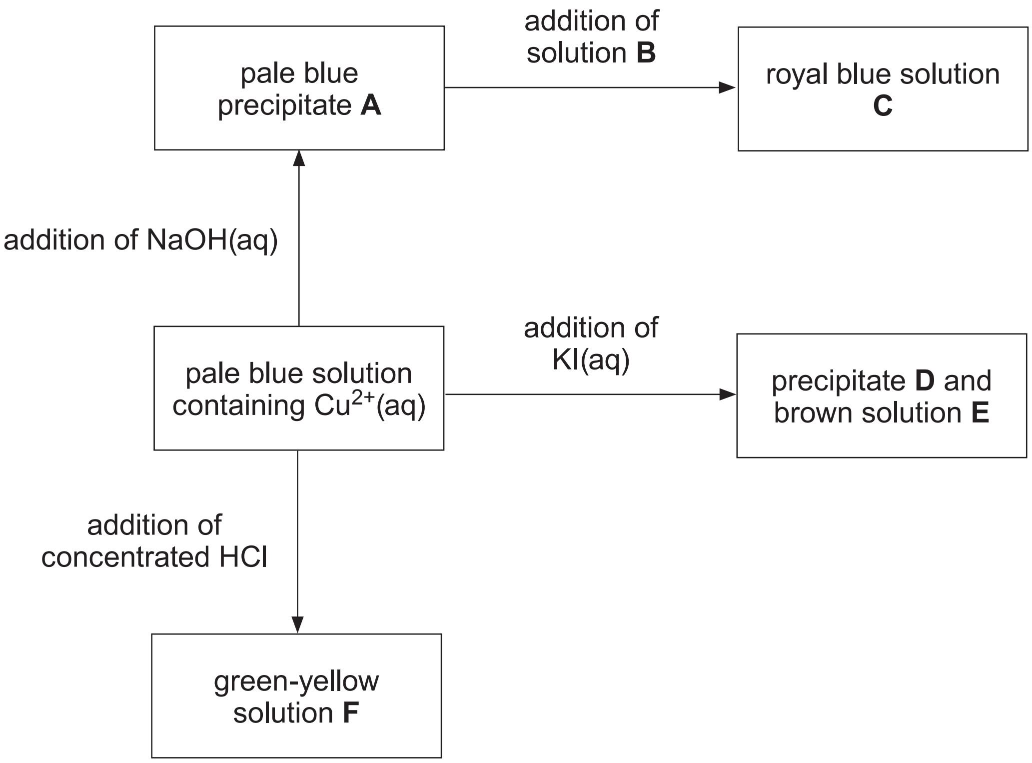
Examiner only

12. *(a)* The diagram below shows some reactions of copper(II) ions in solution.



Identify the species present in each of the following. [3]

A ....

B ....

C ....

D ....

E ....

F ....

Examiner only

(b) Vanadium exhibits the properties of a typical transition element, such as having variable oxidation states and forming coloured complexes. Explain why transition elements have variable oxidation states. [1]

....

....

....

(c) Blue oxovanadium(IV) cations, VO^{2+}, can be oxidised to yellow oxovanadium(V) ions, VO_2^{+}, by bromate(V) ions, BrO_3^{-}.

	Standard electrode potential, E^{θ} / V
$VO_2^{+} + 2H^{+} + e^{-} \rightleftharpoons VO^{2+} + H_2O$	+1.00
$Br_2 + 2e^{-} \rightleftharpoons 2Br^{-}$	+1.09
$BrO_3^{-} + 6H^{+} + 6e^{-} \rightleftharpoons Br^{-} + 3H_2O$	+1.44

Use these standard electrode potentials to identify the main bromine-containing product formed when excess bromate(V) ions are added to a solution containing VO^{2+} ions. Give reasons for your answer. [3]

....

....

....

....

Examiner only

(d) The effect of the concentration of H^+ ions on the rate of the reaction in part *(c)* was studied using the same concentrations and volumes of all other reactants. The following data were collected.

pH	Rate / mol dm^{-3} s^{-1}
1.0	3.287×10^{-3}
1.4	1.308×10^{-3}
1.8	5.193×10^{-4}
2.2	2.071×10^{-4}

Find the order of the reaction with respect to H^+. You **must** show your working. [3]

Order with respect to H^+ = ...

(e) The value of the rate constant, k, for this reaction at two temperatures is given below.

Temperature / K	Rate constant (unit not shown)
288	0.0761
300	0.2873

The activation energy for this reaction is 79.333 kJ mol^{-1} and the Arrhenius equation is:

$$k = Ae^{-\left(\frac{E_a}{RT}\right)}$$

Calculate the value of the rate constant, k, at a temperature of 295 K. [4]

k at 295 K = ...

Examiner only

(f) VO^{2+} ions may also be oxidised to VO_2^{+} ions using coloured Ce^{4+}, which is itself reduced to colourless Ce^{3+}. A titration using this reaction does not usually include an indicator and colorimetry is sometimes used.

State why an indicator may not be useful in this reaction and explain how colorimetry can be used to find the end point of the titration. [3]

....

....

....

....

....

....

17

END OF PAPER